ENGINEERING PORTFOLIO DOSSIER

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Electronic Engineering | Embedded Systems | Edge Intelligence

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B.Eng. Electronic Engineering | CGPA 3.0/4.0 | Expected completion 5 December 2026

SUPPLEMENTARY MATERIAL FOR INTERNATIONAL MSc SCHOLARSHIP AND RESEARCH-SUPERVISION APPLICATIONS

Five project case studies • research direction • implementation boundaries • next validation

01 / RESEARCH IDENTITY

A connected engineering direction

My work is centred on embedded electronic systems that connect sensing, signal conditioning, computation and action. Across the projects, I focus on the complete engineering path: how a physical quantity is acquired, conditioned, converted into digital information, interpreted under resource constraints, communicated to another device or operator, and used to support a reliable decision.

The five selected projects form a progression rather than a collection of unrelated demonstrations. PowerScope develops the instrumentation and acquisition perspective; VibeGuard applies embedded signal features to condition monitoring; GridPulse-RESTORE extends local diagnostics into a feeder-protection and supervised-restoration setting; the ESP32-C3 Environmental Logger and STM32F103 Blue Pill Board develop PCB and firmware-test experience.

Unifying research theme Resource-aware embedded anomaly detection and trustworthy telemetry for electrical and industrial assets.

Five selected projects

Project	Research / engineering contribution	Current evidence boundary
PowerScope	Embedded instrumentation, dual-channel acquisition, FPGA capture, power conversion and system integration.	Final-year team project; design report and repository are available. Do not imply unreported calibration or accuracy results.
GridPulse-RESTORE	Distribution feeder protection, fault isolation, alternate-source assessment and human-approved dead transfer.	A1/A2 drawn, B1 partial, B2/C1/C2 design-only; full closed-loop simulation not yet recorded.
VibeGuard	STM32 vibration features, ADXL345 interface, UART telemetry and Python dashboard.	Prototype with documented Proteus sensor / ADC / sampling limitations.
ESP32-C3 Environmental Logger	KiCad board development, BOM management, test sketches and PCB revision work.	README does not yet disclose exact sensors, calibration or end-to-end logging details.
STM32F103 Blue Pill Board	KiCad microcontroller-board design and schematic-to-PCB workflow.	PCB design study; no public manufacturing / electrical validation report.

Postgraduate areas of interest

- Embedded systems and electronic engineering
- Digital signal processing and embedded instrumentation
- Resource-aware TinyML / edge intelligence
- Condition monitoring and fault diagnostics
- Cyber-physical power systems, control and dependable telemetry

02 / PROJECT CASE STUDY 01

PowerScope

Portable dual-channel oscilloscope & programmable DC power supply

Project context Undergraduate final-year team project at FUTO, 2025–2026. The report names a four-person team. My CV records my role as Lead Hardware & Firmware Designer; this should be presented as collaborative work.

Electronics laboratory work commonly requires a controllable power source and a separate measurement instrument. PowerScope is designed to combine a programmable low-voltage DC supply and a two-channel digital

storage oscilloscope in one portable platform, enabling a user to power a circuit and observe its response within a coordinated instrument.

Documented system architecture

Subsystem	Documented design
Programmable DC supply	1–15 V output, up to 2 A; LM5176 buck-boost controller; LTC4015 battery-management / charging interface.
Oscilloscope ADC	Dual AD9251-20 14-bit ADC architecture; simultaneous two-channel sampling at 10 MS/s per channel as documented in the report.
Analogue front end	Input protection, selectable attenuation and a THS4551 fully differential amplifier per channel to drive the ADC.
Capture and processing	FPGA logic for acquisition timing, channel framing and capture handling.
Supervisory interface	ESP32-S3 recorded in the CV for display telemetry, parameter adjustment and wireless data streaming.
Signal-analysis direction	The CV records an exploratory Edge Impulse classification direction using FFT / DWT feature representations; no industrially validated model performance is claimed here.

My role and evidence

My CV lists Lead Hardware & Firmware Designer. The final-year report, circuit diagrams, calculations and public repository are the primary supporting evidence. The public repository includes oscilloscope firmware and test folders relating to display and ESP32 UART integration.

Limitations and next verification

- Do not advertise calibrated bandwidth, effective ADC resolution, measurement accuracy, noise floor or transient error unless a test report supports the figure.
- Publish a complete block diagram, subsystem schematics and a calibration procedure.
- Compare measurements against a named reference instrument and publish an error table with conditions and uncertainty.
- Separate simulated results from physical measurements and document the FPGA / ESP32 interface clearly.

03 / PROJECT CASE STUDY 02

GridPulse-RESTORE

Fault-aware feeder isolation & human-approved service restoration

Project status Proteus engineering design / partial implementation. A1 and A2 are populated; B1 is partly built; B2, C1 and C2 are specified but were not yet drawn and integrated in the supplied build record. A complete closed-loop simulation had not been run.

Research question

How can a distribution-feeder demonstrator combine local fault protection, isolation, alternate-source eligibility checks and operator-approved restoration without transferring protection authority to a network gateway or paralleling sources?

Architecture and control authority

The plant is an isolated 24 Vrms / 50 Hz model with three radial feeders and a separate alternate source. The STM32F103C8T6 is the local protection authority; the ESP32 forwards status and operator requests only.

The sequence is monitor → detect → isolate → assess alternate-source capacity → request human approval → dead transfer (break-before-make) → verify feedback → restore or block. The sources are never paralleled.

Corrections to preserve

- Controller: STM32F103C8T6, not STM32F407. TR1: 2P5S transformer with four tap-selection switches only; do not add the superseded fifth switch.
- TR2: 100:1 CT in the neutral-earthing path for earth-current measurement, not a feeder-load sensor.
- Surge clamp: back-to-back 43 V 1N4755A zeners. F3 rectifier / capacitor load: corrected 220 Ω resistor, not 33 Ω.
- Retain the original architecture; use dead transfer / break-before-make and never parallel the sources.

Implementation status

Block	Function	Recorded status
A1	Source, transformer, earthing, surge clamp and control power	Drawn / populated
A2	Three feeders, loads, fault points, tie points and feedback	Drawn / populated
B1	Measurement and analogue protection	Partly built
B2	STM32 control core / I/O expansion	Design specified; not integrated in recorded build
C1	ESP32 supervision / operator approval	Design specified; not integrated in recorded build
C2	Test bench / communications and security matrix	Design specified; not integrated in recorded build

Fault currents and trip times are calculated design values, not measured or full-system simulation results. Next: complete the remaining blocks, integrate firmware and run repeatable fault, communications-loss and interlock tests.

04 / PROJECT CASE STUDY 03

VibeGuard

STM32 vibration monitoring & anomaly-detection prototype

VibeGuard is an STM32F103-series / ADXL345 condition-monitoring prototype. C firmware extracts RMS and peak-to-peak features and streams telemetry over UART to a Python Streamlit / Pandas dashboard for plots, anomaly states and CSV logging.

Signal and software path

Signal path: ADXL345 readings → STM32 feature extraction → UART / virtual COM-port bridge → Python dashboard. A 555 timer and 74HC4017 counter provide a separate visual boot sequence in the Proteus setup.

Schematic corrections retained

- ADXL345: separate SDO from SDA and tie SDO to GND (address 0x53); SDA → PB9, SCL → PB8.
- I^2C pull-ups: 4.7 kΩ to 3.3 V; check for 5 V pull-ups from any LCD backpack and level-shift if needed.
- UART: PA9 / TX → terminal RXD; PA10 / RX ← terminal TXD; common ground.
- Vibration input: PA0 only after conditioning to 0–3.3 V; bias any bipolar waveform.
- Retain the 10 kΩ NRST pull-up, BOOT0 pull-down and local decoupling checks.

Documented simulation limitations

The repository README labels it a prototype. In the documented Proteus setup, the ADXL345 is control-panel driven rather than physically excited; the hardware I^2C model did not respond; and the ADC returned zero in Proteus 8.13, so software I^2C was used. Effective sampling is about 40 Hz (below 20 Hz content). The 29 mg / 46 mg thresholds are demo mappings, not ISO 20816-compliant measurements.

Next validation work

- Move to a validated hardware I^2C / SPI path and sensor FIFO; test with known signals.
- Add band-limited acceleration-to-velocity processing and FFT features.

- Measure false alarms, detection quality, latency and memory on labelled cases.
- Repeat key tests on physical hardware before real-machine or standards-compliance claims.

Demo media A dedicated website video player sits directly below VibeGuard. The actual recording was not supplied; add an MP4 or YouTube URL in content.js to activate it.

05 / PROJECT CASE STUDIES 04 AND 05

ESP32-C3 Environmental Logger

Environmental logging platform — PCB and firmware test work

The public repository contains a KiCad project, test sketches, BOM.ods, BOM11.xlsx and an image documenting improvements to PCB prototype 2. This supports a description centred on board development, component planning, firmware tests and iterative PCB revision.

Evidence boundary The public README is currently empty: exact sensor part numbers, measurement ranges, calibration, sample interval, data format and logging destination are not established and should not be guessed.

Next evidence to publish

- Add a system block diagram and exact ESP32-C3 board variant.
- List sensor part numbers, pin map, power budget and units.
- Document firmware dependencies, log format and sample data.
- Add PCB revision notes, assembled-board images and ERC / DRC reports.

STM32F103 Blue Pill Board

KiCad-based microcontroller board design study

The public README describes a basic STM32 Blue Pill-based KiCad board design, referencing Phil's Lab STM32 PCB design tutorial. It is a board-design study covering schematic organisation, component placement, routing and project-file preparation.

Evidence boundary Describe this as a PCB design study. The public repository does not currently provide a detailed electrical test report, manufactured-board evidence or production validation.

Next evidence to publish

- Publish readable schematic and front / back PCB renders.
- Add the BOM, board dimensions, ERC / DRC report and manufacturing outputs.
- Document power rails, clock source, SWD access and connector pinout.
- Record continuity, power-up and programming tests if performed.

Research questions for postgraduate work

Theme	Potential research question	Methods / evidence
Resource-aware edge diagnostics	How can a small MCU distinguish abnormal operating conditions under compute, memory, energy and sampling constraints?	Threshold / statistical baseline; compact ML comparison; latency, memory and energy profiling.
Embedded instrumentation	How should signal conditioning, conversion, sampling, calibration and digital processing be partitioned in compact instruments?	Analogue front end; ADC architecture; DSP; reference-instrument comparison; uncertainty.
Cyber-physical power systems	How can local protection and supervisory control coordinate restoration without weakening independent protection authority?	Feeder modelling; protection interlocks; event state machines; supervised restoration tests.
Trustworthy telemetry	How can sensing systems preserve traceable measurements when data quality varies or links fail?	Embedded networks; event logs; protocol framing; failure handling; provenance.

Preprints recorded in the academic CV

Okoro, C. N. (October 2026). “A Human-on-the-Loop Cyber-Physical Framework for Fault-Aware Service Restoration in Nigerian Distribution Feeders: Requirements Derived from EEDC Industrial Training.” Engineering Archive preprint. DOI: https://doi.org/10.31224/8498

“PowerScope: Preliminary Design of a Portable Dual-Channel Digital Oscilloscope and Programmable DC Power Supply.” Collaborative preliminary-design preprint. DOI: https://doi.org/10.31224/8499

Publication status These entries are preprints, not peer-reviewed journal publications.

Research funding application — ACE-FUELS (FUTO)

Application in progress (not yet awarded). I am seeking research-grant support from the African Center of Excellence in Future Energies and Electrochemical Systems (ACE-FUELS), domiciled at FUTO, to further develop my first-author Human-on-the-Loop cyber-physical framework for fault-aware service restoration in Nigerian distribution feeders. The proposed continuation focuses on strengthening the restoration model and staged simulation/validation protocol. The potential connection to ACE-FUELS’ Clean and Future Energy remit is my proposed interdisciplinary research fit, not an existing centre partnership or endorsement. *Official centre profile: https://acefuels-futo.org/about-us/*

Academic preparation and credentials

- B.Eng. Electronic Engineering, Federal University of Technology, Owerri (FUTO) — CGPA 3.0/4.0; expected completion 5 December 2026.
- Arm Cortex-M Processors Overview — Arm Education / Coursera; verification: https://coursera.org/verify/XN6QVOA7JFD3.
- Altium Designer Curriculum & Cumulative Examination — Altium Education (listed in CV; verification URL not supplied).
- MATLAB Onramp and Simulink Onramp — MathWorks (listed in CV).
- Embedded Systems Course Completion — SkillUp by Simplilearn (listed in CV).
- Real-Time Operating Systems for Beginners — MindLuster; 1 hour; 13 June 2026; certificate no. 925934e4.
- Embedded C Programming Course for Beginners — Cursa; 17 h 08 min; 1–4 October 2026; validation code 546269ede3e3a628628d3149435e1b0a.
- Basic Radio Awareness — Tait Radio Academy (listed in CV).
- Professional affiliations listed in CV: IEEE Student Member, IEEE Signal Processing Society and IEEE Power Electronics Society.

Experience beyond the project bench

EEDC / MainPower — Engineering SIWES

The uploaded completion letter confirms a six-month industrial attachment and work areas including Operations, Safety and Administration (Stores). The academic CV records exposure to right-of-way and substation safety, transformer / switchgear inventory, distribution expansion support, load-profile work, transformer insulation and earth-resistance testing, power-quality troubleshooting and distribution maintenance. The portfolio includes the completion letter as a supporting document.

FRCN / Tait Radio Academy — Telecommunications internship

The CV records a five-month attachment covering RF and broadcast systems, VHF / UHF propagation, transmitter maintenance, repeater-based communications, analogue FM and digital trunked-radio concepts. It also records completion of Basic Radio Awareness learning through Tait Radio Academy.

Departmental academic support and leadership

The current CV records Academic Assistant service in FUTO's Petroleum Engineering programme (100 Level, second semester), under class advisor Dr. Nnanna Okoli (a different department to avoid grading bias), supporting assessment grading and confidential result records.

UAV engine / motor-monitoring team project

The CV records leadership of a nine-member team developing a microcontroller-based monitoring concept for temperature, RPM, fuel-flow and piezoelectric vibration signals. It describes moving-average / Kalman-filter preprocessing and an SVM classification approach. Quantitative performance figures are intentionally omitted here because the supplied public portfolio does not include the dataset, evaluation protocol or reproducible test record needed to substantiate them.

Technical toolkit

Area	Tools / methods appearing in source materials
Embedded platforms	STM32 / ARM Cortex-M; ESP32 / ESP32-S3 / ESP32-C3; ATmega / AVR; PIC.
Firmware / interfaces	Embedded C; C++; UART; I^2C; SPI; GPIO; ADC; peripheral integration.
Electronics / PCB	KiCad; Altium Designer; Proteus VSM; LTspice; schematic capture; PCB layout; BOM preparation.
Signal / edge	RMS and peak-to-peak features; FFT / DWT concepts; sensor acquisition; anomaly thresholds; Edge Impulse.
Software / modelling	Python; Streamlit; Pandas; MATLAB; Simulink; PlatformIO; VS Code; Git / GitHub; LaTeX.

Evidence principle This is a summary of tools and methods in the source materials, not a proficiency ranking or a claim of expert-level mastery across every item.

Portfolio evidence checklist

- Repository and dated commit; problem statement, role and block diagram.
- Source files, schematics / PCB files and reproducible setup steps.
- Test procedure with units, test count, expected behaviour and recorded outcomes.
- Screenshots / video clearly labelled as simulation or physical hardware.
- Limitations, unresolved issues and next experiment.

For scholarship review, the objective is to make each project inspectable and discussable. Clear separation between implemented work, design proposals, simulation results and physical measurements makes the portfolio more credible and useful to a prospective supervisor.